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I'm a ROBO SCIENTIST

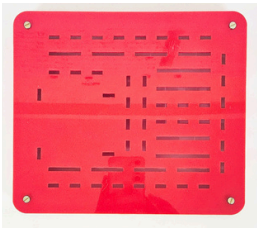
SESSION NAME

Session

8

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1



2



3



4



5



6



7



8



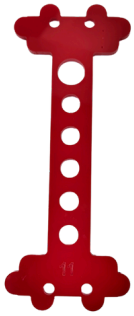
9



10



11



1



Spacers

2



L Clamp

3



Spacer Screws

4



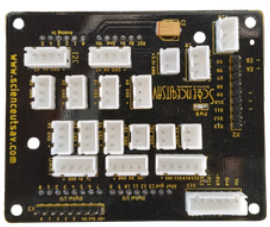
Nuts

5



Screw

6



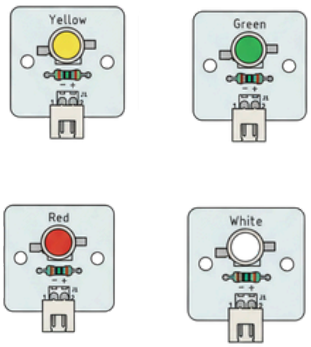
Arduino With Shield

7



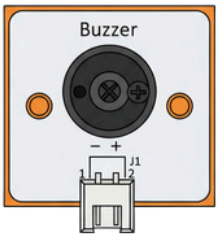
Arduino cable

ELECTRONIC COMPONENTS



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



Buzzer

A buzzer is an electronic sound-producing device that generates audible tones or beeps when powered. It is commonly used in alarms, notifications, indicators, security systems, and electronic projects to provide sound-based alerts and feedback.



Push Button

A push button is a simple mechanical switch that controls an electrical circuit when pressed. It is widely used in electronic devices, control panels, automation systems, and DIY projects to provide user input and trigger specific actions.



LDR Sensor (Light Dependent Resistor)

An LDR sensor is a light-sensitive electronic component that changes its resistance according to the intensity of light falling on it. It is commonly used in automatic lighting systems, light detection circuits, security devices, and electronic projects to sense and respond to changes in ambient light.

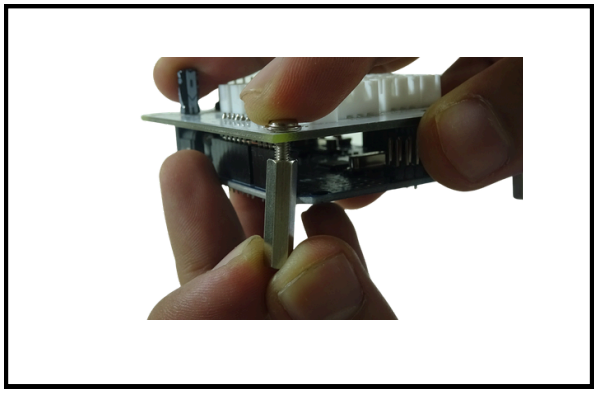


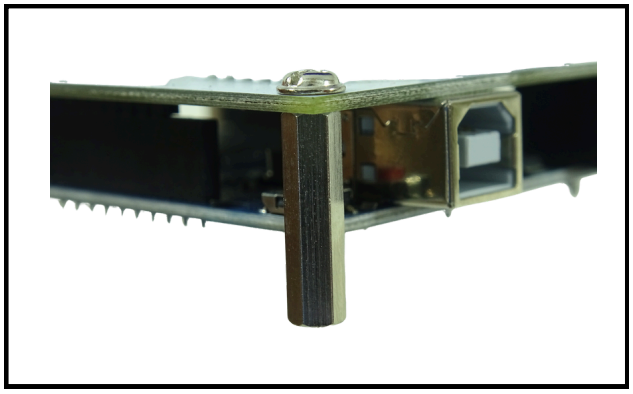
2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

Procedure :

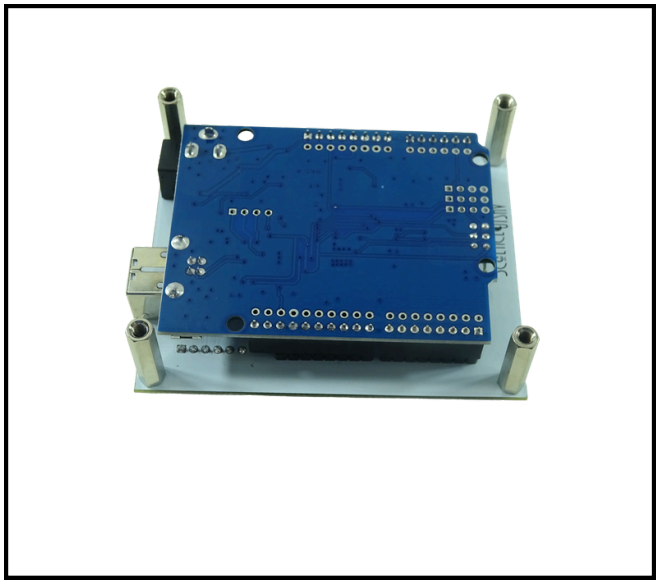
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

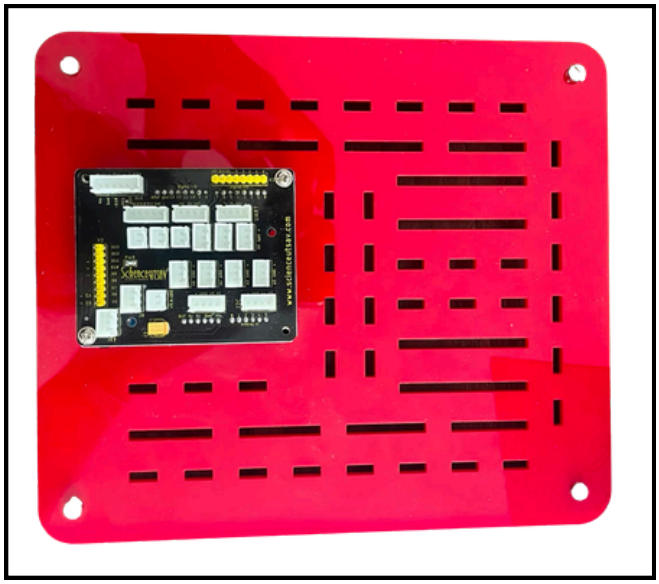




Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield

Step 3 : Place the Arduino Shield to the base board as shown below



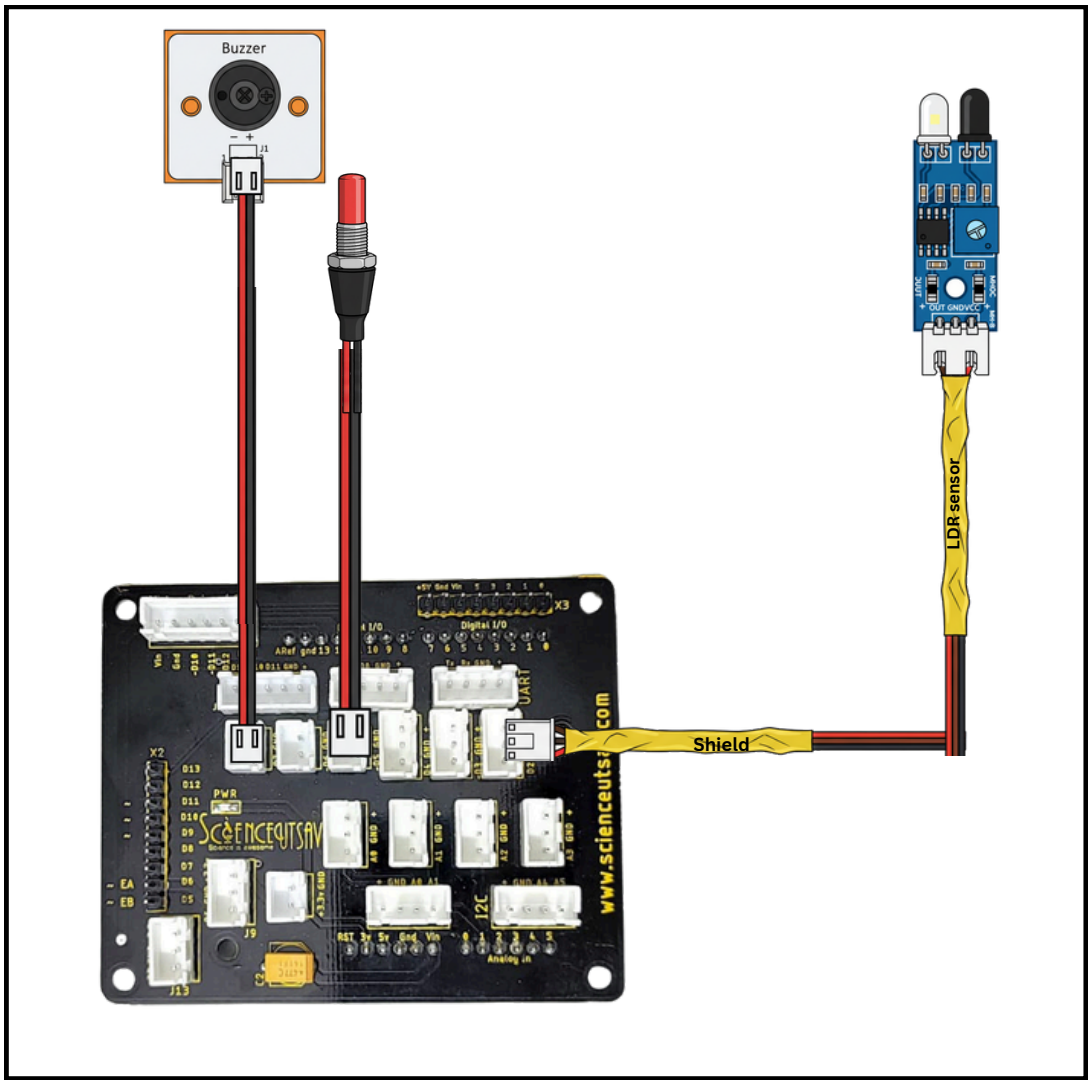


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

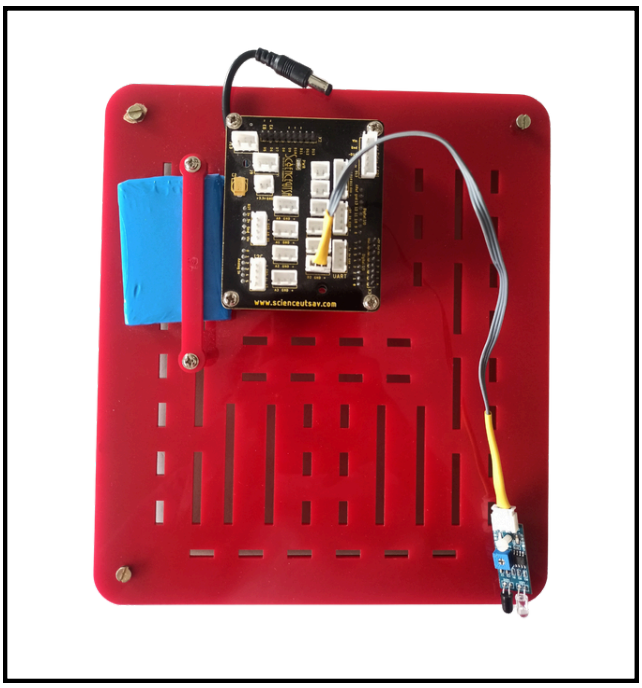
The circuit diagram of the project is given below



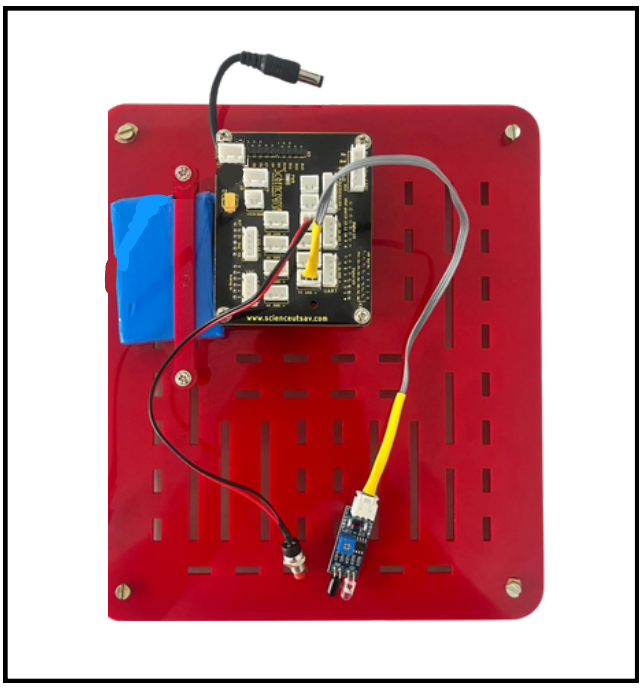
CONNECTION:

1. IR sensor- D2
2. Push button- D5
3. Buzzer- D7

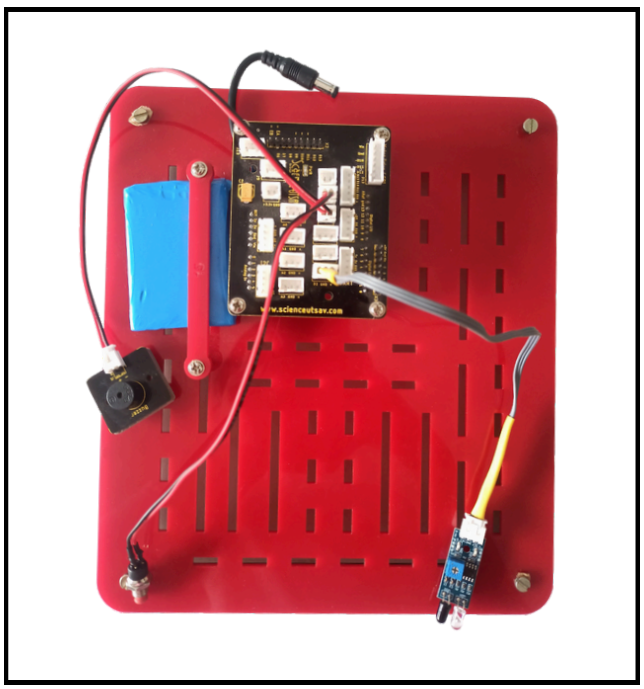
Step 1- Connect the IR sensor to the shield



Step 2- Attach the push button to the shield as shown



Step 3- Attach the buzzer as shown



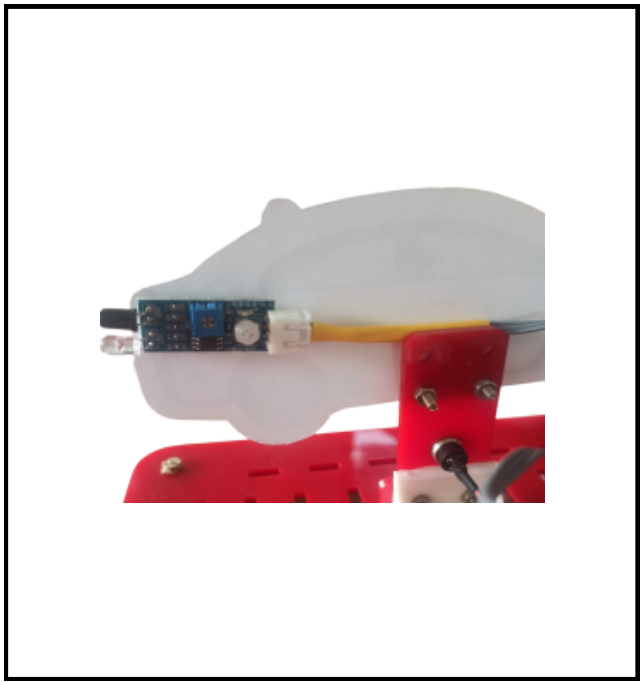
Step 4- Fix the template to the stand

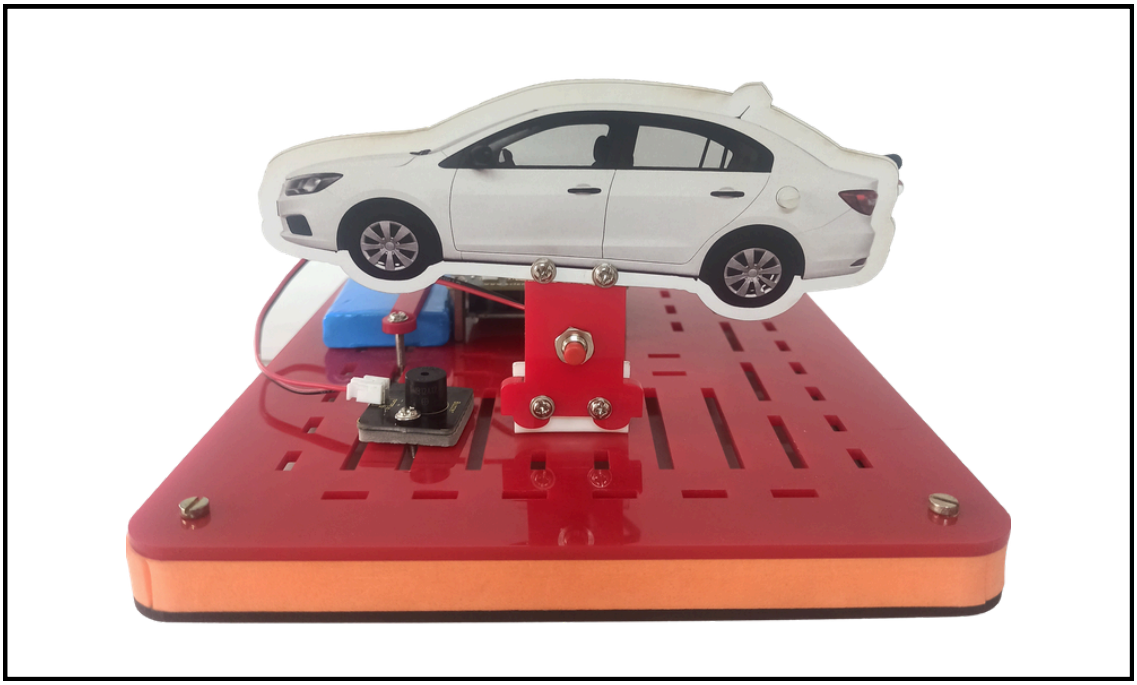


Step 5- Fix the stand to the board



Step 6- Attach the IR sesor to the back of the template and fix the push button





Code

```
// Session 8 Grade 6
// Car Back Safety System
// IR detects object = buzzer ON
// Push button turns alarm OFF until object is removed

const int irSensor = 2;
const int pushButton = 5;
const int buzzer = 7;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

bool alarmMuted = false;

void setup() {
  pinMode(irSensor, INPUT);
  pinMode(pushButton, INPUT_PULLUP);
  pinMode(buzzer, OUTPUT);
}

void loop() {
  bool objectDetected = digitalRead(irSensor) == IR_DETECTED;

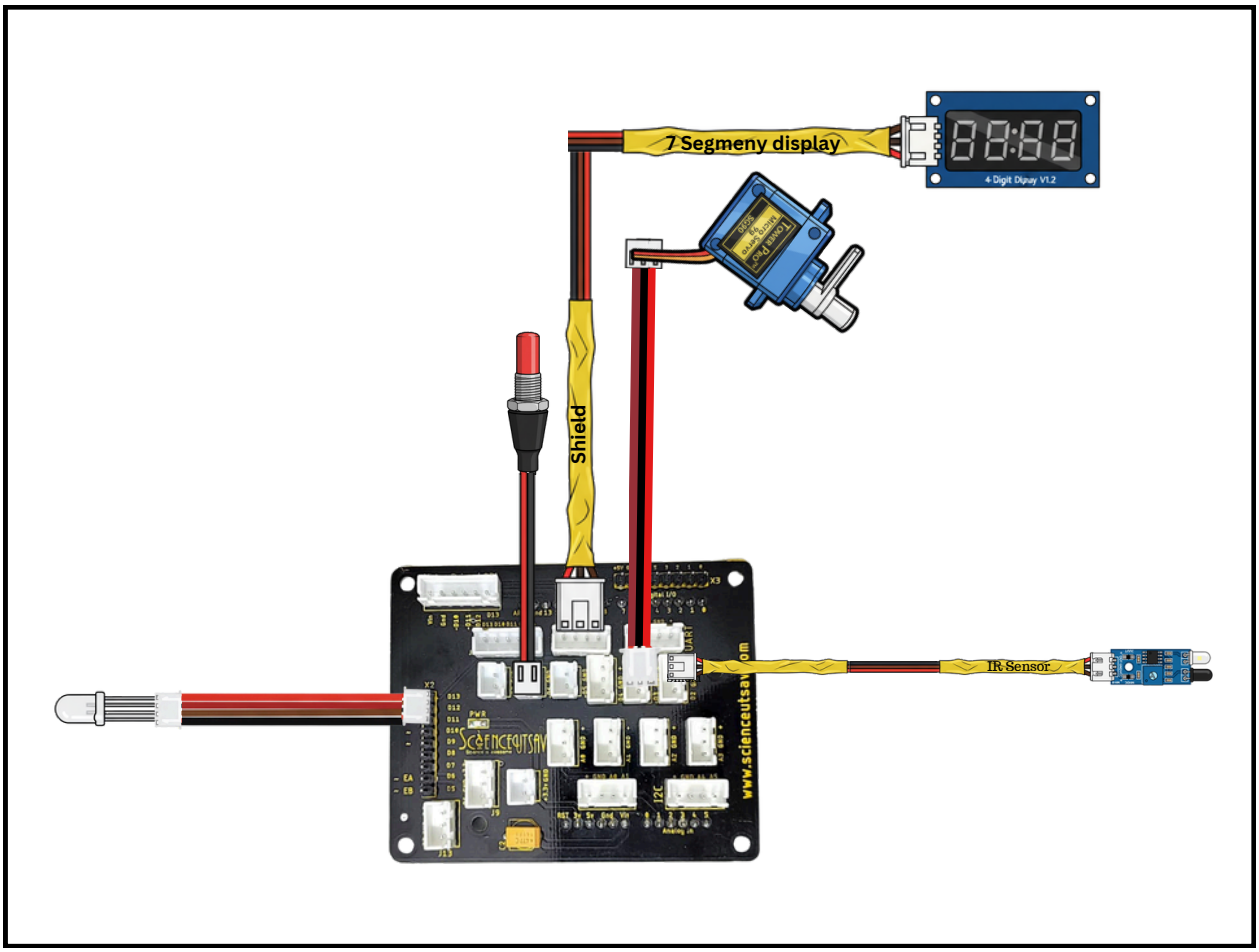
  if (digitalRead(pushButton) == LOW) {
    alarmMuted = true;
    digitalWrite(buzzer, LOW);
    delay(300);
  }

  if (!objectDetected) {
    alarmMuted = false;
  }

  if (objectDetected && !alarmMuted) {
    digitalWrite(buzzer, HIGH);
  } else {
    digitalWrite(buzzer, LOW);
  }
}
```

INTERMEDIATE PROJECT

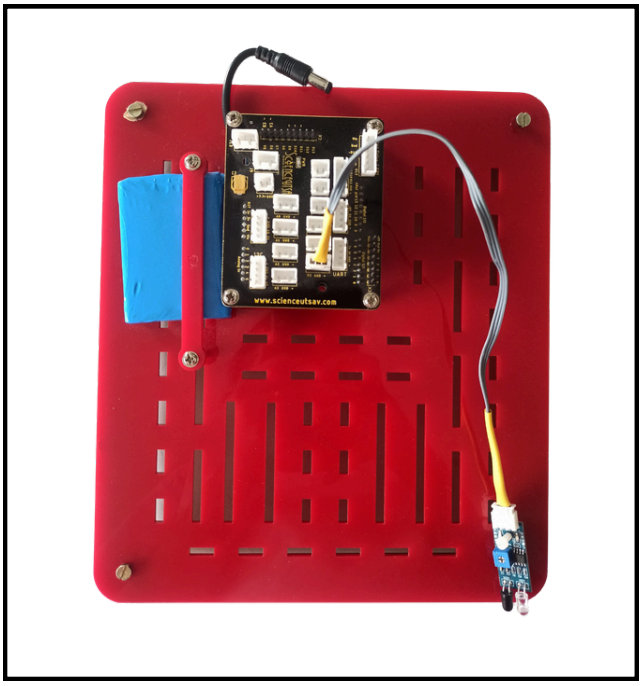
The circuit diagram of the project is given below



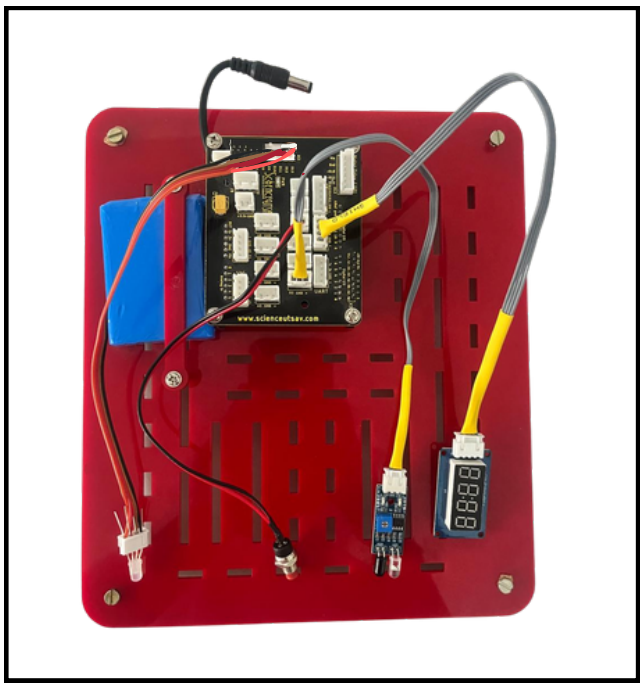
CONNECTION:

1. IR sensor- D2
2. Push button- D5
3. 7-segment display- D8, D9
4. RGB LED-
5. Servomotor- D3

Step 1- Connect the IR sensor to the shield



Step 2- Attach the other components as shown



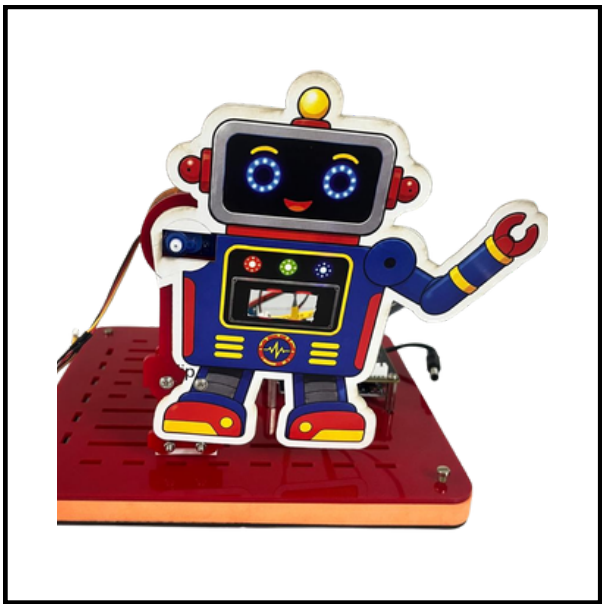
Step 3- Fix the stands as shown and attach the servomotor to it



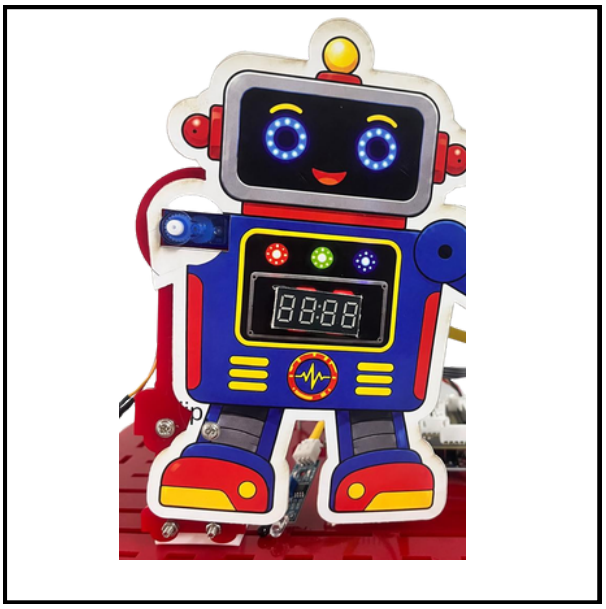
Step 4- Attach the stand to the back of the template as shown



Step 5- Attach the servomotor to the shield



Step 6- Fix the display onto the template



Step 7- Fix the push button as shown



Step 8- Attach the RGB LED to the top of the template





Code

```
// Session 8 Grade 7
// Dancing Robot
// IR detects person = RGB and 7 segment dance pattern starts
// Push button turns it OFF
// D12 is kept always HIGH for RGB project

#include <TM1637Display.h>

const int irSensor = 2;
const int pushButton = 5;
const int redPin = 10;
const int greenPin = 11;
const int bluePin = 12;
const int CLK = 9;
const int DIO = 8;

const int IR_DETECTED = LOW;   // change to HIGH if opposite

TM1637Display display(CLK, DIO);

bool danceOff = false;
int patternNumber = 0;

void setup() {
  pinMode(irSensor, INPUT);
  pinMode(pushButton, INPUT_PULLUP);
  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);

  digitalWrite(bluePin, HIGH); // D12 always HIGH

  display.setBrightness(7);
  display.showNumberDec(0, true);
}
void loop() {
 digitalWrite(bluePin, HIGH); // keep D12 always HIGH
```

```
  if (digitalRead(pushButton) == LOW) {
    danceOff = true;
    allOffExceptD12();
    display.showNumberDec(0, true);
    delay(300);
  }

  if (digitalRead(irSensor) == IR_DETECTED) {
    danceOff = false;
  }

  if (digitalRead(irSensor) == IR_DETECTED && !danceOff) {
    dancePattern();
  } else {
    allOffExceptD12();
  }
}
void dancePattern() {
 patternNumber++;
 if (patternNumber > 9) patternNumber = 1;

 display.showNumberDec(patternNumber, true);

 digitalWrite(redPin, HIGH);
 digitalWrite(greenPin, LOW);
 digitalWrite(bluePin, HIGH);
 delay(200);

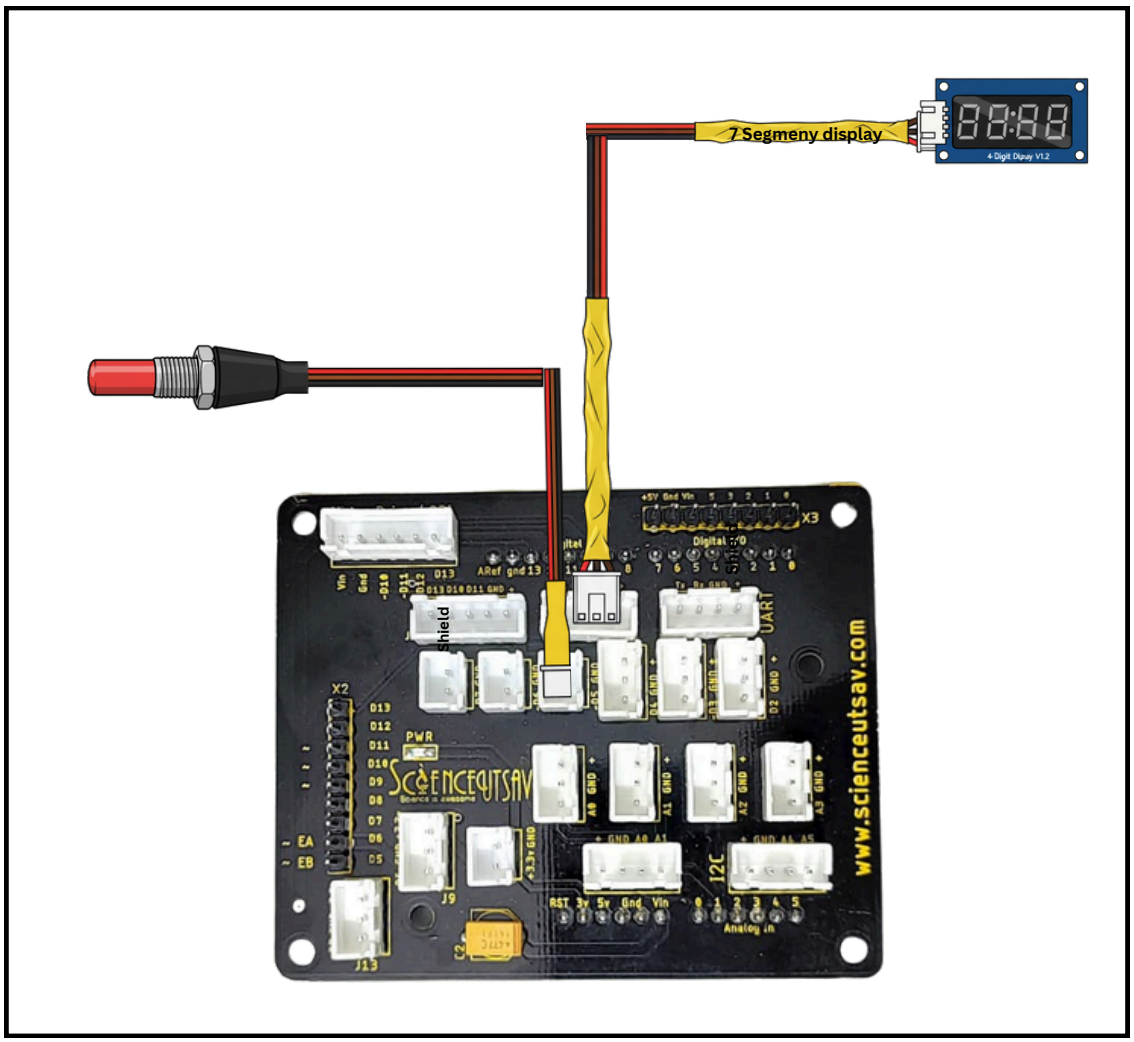
 digitalWrite(redPin, LOW);
 digitalWrite(greenPin, HIGH);
 digitalWrite(bluePin, HIGH);
 delay(200);

 digitalWrite(redPin, HIGH);
 digitalWrite(greenPin, HIGH);
 digitalWrite(bluePin, HIGH);
 delay(200);
}

void allOffExceptD12() {
 digitalWrite(redPin, LOW);
 digitalWrite(greenPin, LOW);
 digitalWrite(bluePin, HIGH);
}
```

ADVANCE PROJECT

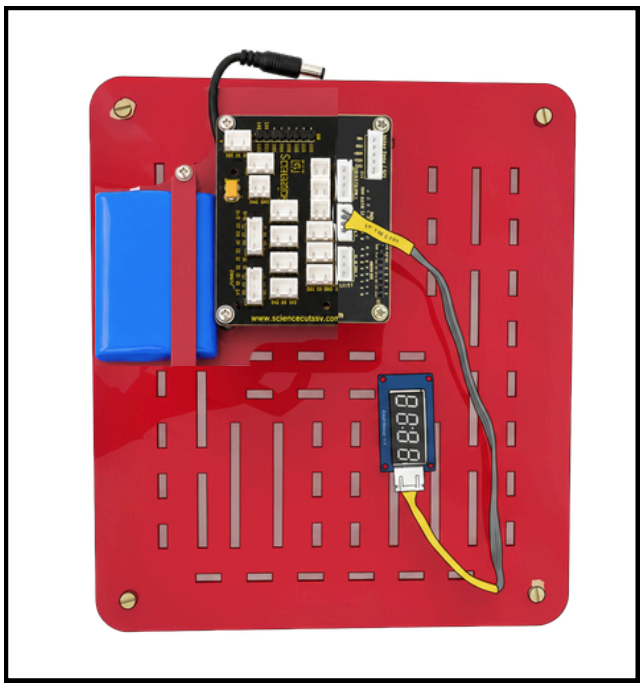
The circuit diagram of the project is given below



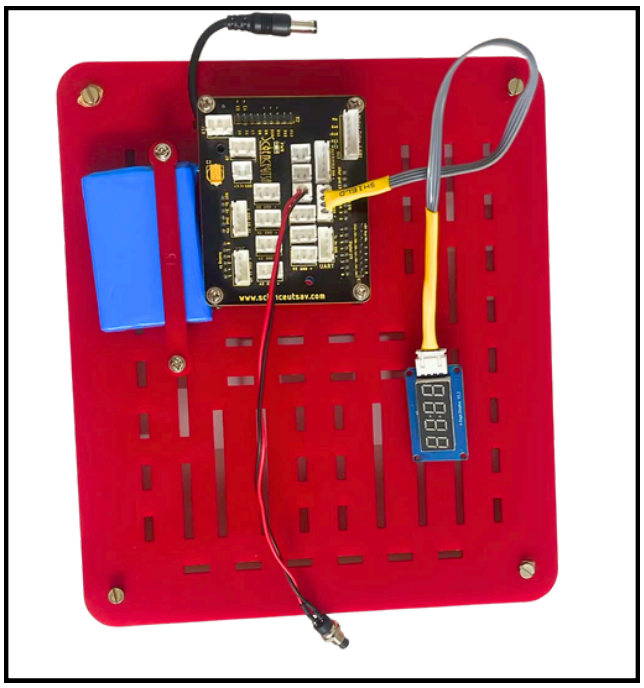
CONNECTION:

1. 7-segment display- D8, D9
2. Push button- D5

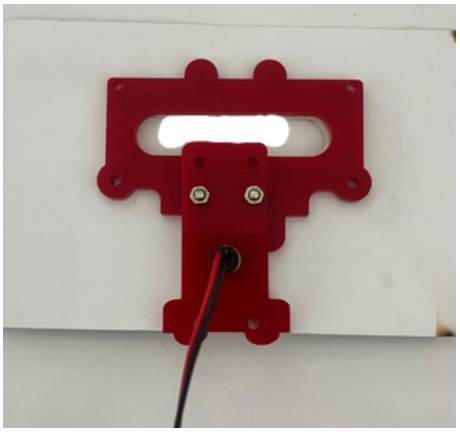
Step 1- Connect the display to the shield



Step 2- Attach the push button

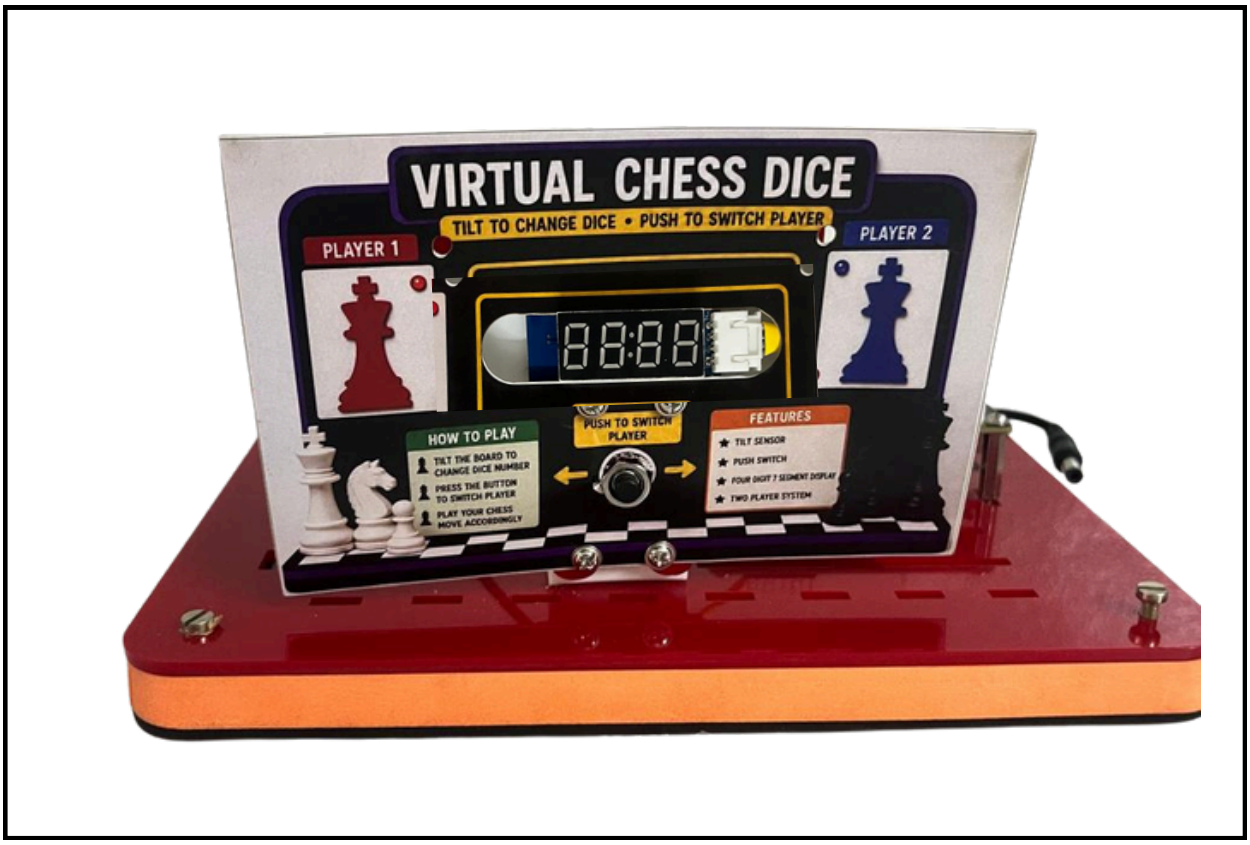


Step 3- Fix the stands as shown and attach them to the template



Step 4- Insert the display through the template




VIRTUAL CHESS DICE
TILT TO CHANGE DICE • PUSH TO SWITCH PLAYER
PLAYER 1
PLAYER 2
HOW TO PLAY
TILT THE BOARD TO CHANGE DICE NUMBER
PRESS THE BUTTON TO SWITCH PLAYER
PLAY YOUR CHESS MOVE ACCORDINGLY
PUSH TO SWITCH PLAYER
FEATURES
TILT SENSOR
PUSH SWITCH
FOUR DIGIT 7 SEGMENT DISPLAY
TWO PLAYER SYSTEM

Code

```
// Session 8 Grade 8
// Virtual Dice
// Tilt sensor rolls dice
// Push button selects player
// Display format: player number on left, dice number on right

#include <TM1637Display.h>

const int tiltSensor = 2;
const int pushButton = 5;
const int CLK = 9;
const int DIO = 8;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

TM1637Display display(CLK, DIO);

int player = 1;
int diceValue = 1;
bool lastTilt = LOW;
bool lastButton = HIGH;

void setup() {
  pinMode(tiltSensor, INPUT);
  pinMode(pushButton, INPUT_PULLUP);
  display.setBrightness(7);
  showPlayerDice();
  randomSeed(analogRead(A1));
}
```

```
void loop() {
  bool tiltNow = digitalRead(tiltSensor);
  bool buttonNow = digitalRead(pushButton);

  if (buttonNow == LOW && lastButton == HIGH) {
    player++;
    if (player > 4) player = 1;
    showPlayerDice();
    delay(300);
  }

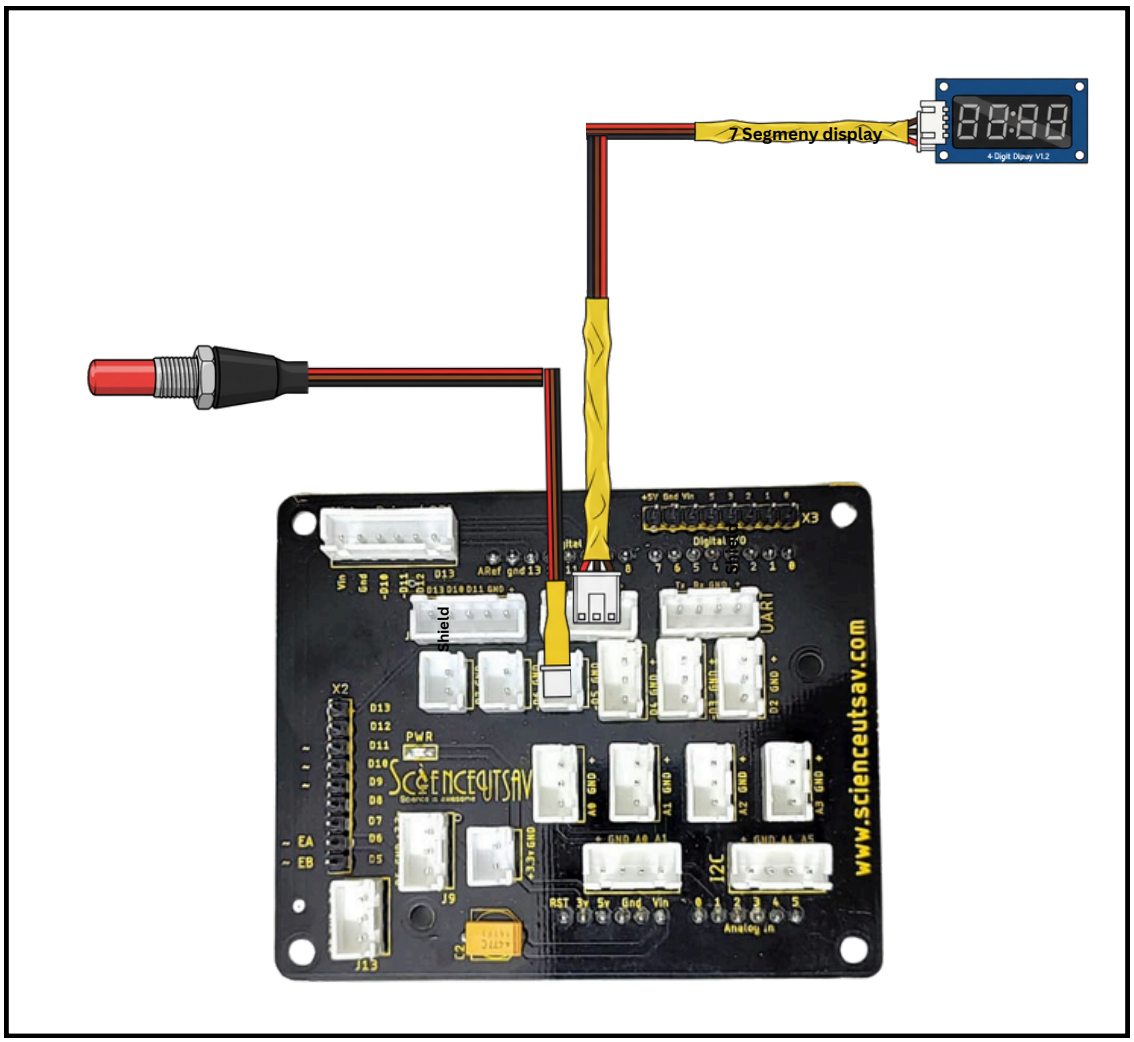
  if (tiltNow == TILT_ACTIVE && lastTilt != TILT_ACTIVE) {
    diceValue = random(1, 7);
    showPlayerDice();
    delay(500);
  }

  lastTilt = tiltNow;
  lastButton = buttonNow;
}

void showPlayerDice() {
  int value = player * 100 + diceValue;
  display.showNumberDec(value, true);
}
```

EXPERT PROJECT

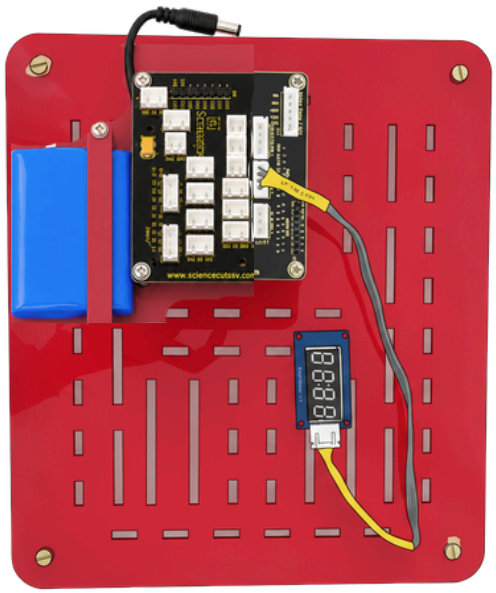
The circuit diagram of the project is given below



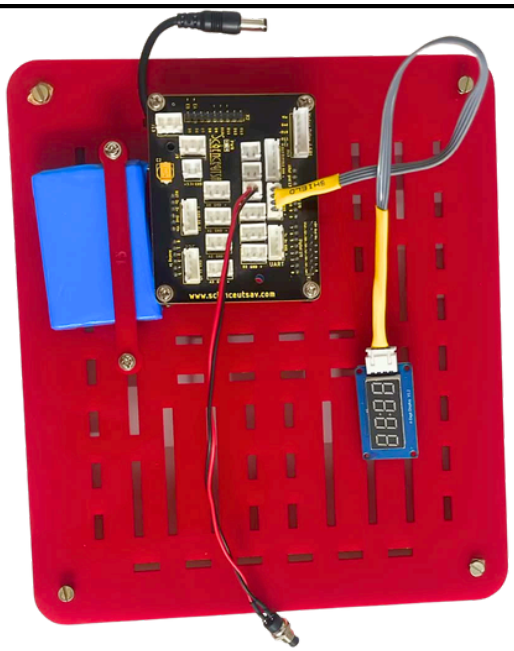
CONNECTION:

1. 7-segment display- D8, D9
2. Push button- D5

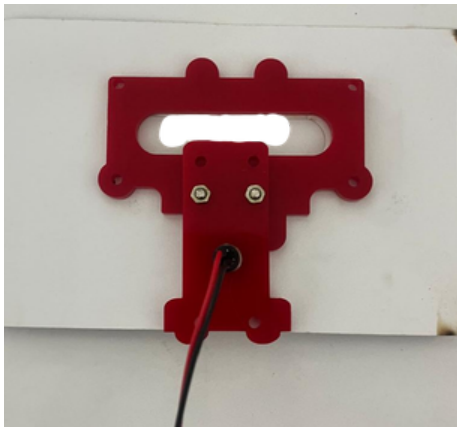
Step 1- Connect the display to the shield



Step 2- Attach the push button

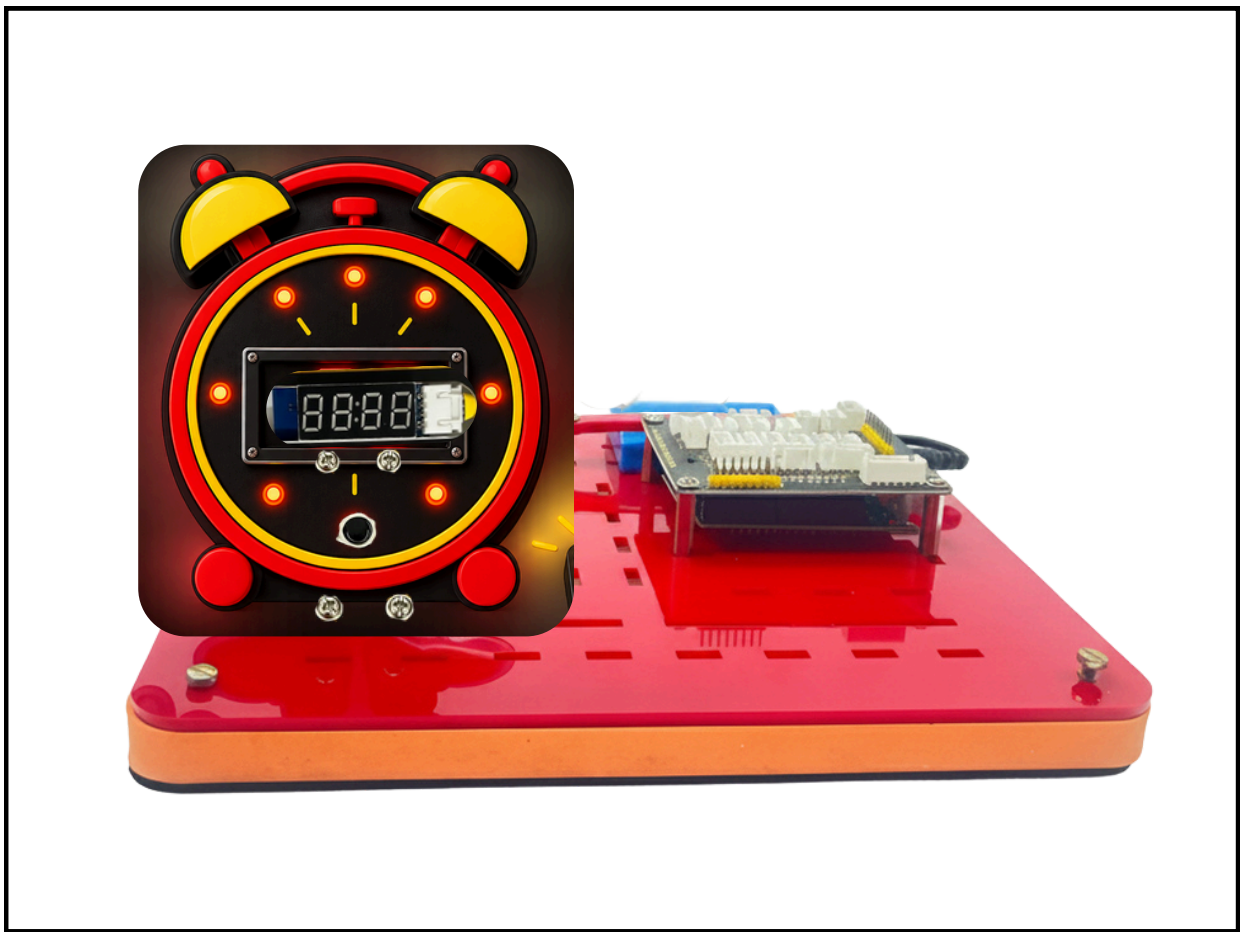


Step 3- Fix the stands as shown and attach them to the template



Step 4- Insert the display through the template





Code

```
// Session 8 Grade 9
// LDR Alarm Timer
// LDR detects light/dark change and timer starts
// Push button turns off / resets the timer

#include <TM1637Display.h>

const int ldrSensor = A0;
const int pushButton = 5;
const int CLK = 9;
const int DIO = 8;

int darkValue = 500; // adjust based on room light

TM1637Display display(CLK, DIO);

bool timerRunning = false;
unsigned long startTime = 0;

void setup() {
  pinMode(pushButton, INPUT_PULLUP);
  display.setBrightness(7);
  display.showNumberDec(0, true);
}
```

```
void loop() {
  int ldrValue = analogRead(ldrSensor);

  if (ldrValue > darkValue && !timerRunning) {
    timerRunning = true;
    startTime = millis();
  }

  if (digitalRead(pushButton) == LOW) {
    timerRunning = false;
    display.showNumberDec(0, true);
    delay(300);
  }

  if (timerRunning) {
    unsigned long seconds = (millis() - startTime) / 1000;
    if (seconds > 9999) seconds = 9999;
    display.showNumberDec(seconds, true);
  }
}
```